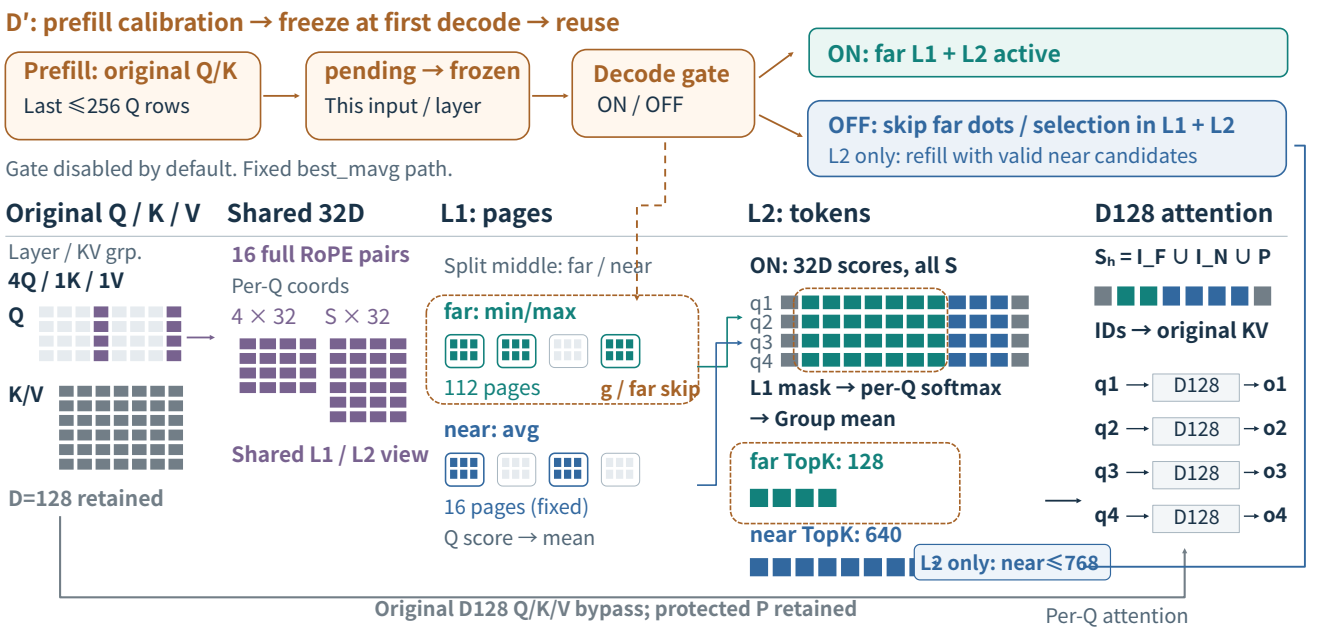

D′: prefill calibration → freeze at first decode → reuse
Prefill: original Q/K
Last ⩽256 Q rows
pending → frozen
This input / layer
Decode gate
ON / OFF
ON: far L1 + L2 active
OFF: skip far dots / selection in L1 + L2
L2 only: refill with valid near candidates
Gate disabled by default. Fixed best_mavg path.
Original Q / K / V
Shared 32D
L1: pages
L2: tokens
D128 attention
Layer / KV grp.
4Q / 1K / 1V
Q
K/V
D=128 retained
16 full RoPE pairs
Per-Q coords
4 × 32
S × 32
Shared L1 / L2 view
Split middle: far / near
far: min/max
112 pages
g / far skip
near: avg
16 pages (fixed)
Q score → mean
ON: 32D scores, all S
q1
q2
q3
q4
L1 mask → per-Q softmax
→ Group mean
far TopK: 128
near TopK: 640
L2 only: near⩽768
Sh = I_F ∪ I_N ∪ P
IDs → original KV
q1 → D128 → o1
q2 → D128 → o2
q3 → D128 → o3
q4 → D128 → o4
Original D128 Q/K/V bypass; protected P retained
Per-Q attention